

Worksheet 0

An echo of C

No Marks are awarded for this worksheet, however, all tasks MUST be completed before preceding.

All work is handed in via git/GitLab and will require the skills demonstrated in this worksheet.

Compiling C with clang

Before we begin it is important to be able to compile C++ programs, which you should already be confident at, but in case you are feeling rusty we compile a simple hello world program to get started. Assuming you are logged into the remote server, create a directory to work in and create the a C++ file (hello-world.cpp) with the following content:

```
#include <stdio.h>

int main(void) {

    printf("Hello World\n");

    return 0;
}
```

This can be compiled with the following:

```
clang -o hello hello-world.cpp
```

and will result in the executable hello.

Basic Pointers

Task 1: Create a pointer to the local variable `n` called `ptr_to_n`, and use it to increase the value of `n` by one. Print the result.

Create an array that has 3 elements (10, 30, 2000), and declare a pointer to that array, and then a loop that iterates through the array, using the pointer, printing each element, and the pointer of that element.

Implement a function that takes two pointers, to the same type, along with a length parameter. If the pointers are not null, then check that the elements are the same, for length, and return true (1), otherwise false (0). Write a main function to test your function.

Write a program that reads the following numbers from a file, and prints their sum.

foo.txt

```
10
20
30
40
54
```

Implement a program that contains the swap function, and tests it works correctly.

```
void swap(void * x, void * y);
```

Implement a function to prints a 2D-array, given a pointer to the array, and its corresponding width and height.

```
void print_array(int * arr, int width, int height);
```

Once you have completed commit your work to git and then demonstrate to your lab tutor for feedback.

TicTacToe

To complete this worksheet you should have look at implementing TicTacToe. Again push your resulting implementation to Gitlab.

Your program:

- multiple source files (.c and .h);
- a function that uses pointers

This should be a fairly straightforward task and we should aim to finish it ASAP.